

Methods 3: Multilevel Statistical Modeling and Machine Learning

Week 41: *Explanation versus Prediction*

October 7, 2025

The course plan

Week 37:

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model (Lau)

Week 38:

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects? (Lydia)

Week 39:

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous? (Lau)

Week 40:

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review (Lydia)

Week 41:

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models (Lydia)

Week 42:

no teaching (go fetch potatoes instead)

Week 43:

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running (Lau)

Week 44:

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses (Lau)

Week 45:

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review (Lydia: To be moved)

Week 46:

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data? (Lau)

Week 47:

Lesson 9: Neural networks

Class 9: Code review (Lau)

Week 48:

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource) (Lydia)

Week 49:

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning (Lydia)

Week 50:

Student presentations:

Class 12: Work on portfolios: Ask anything! (Lydia & Lau)

Likelihood confusion

$$L(\hat{\beta}|y) = \prod_{i=1}^n \hat{y}_i^{y_i} (1 - \hat{y}_i)^{(1-y_i)}$$

$$\hat{\beta} \mapsto f(y \mid \hat{\beta}) = \mathcal{L}(\hat{\beta} \mid y)$$

Recap

EVALUATING GENERALISED LINEAR MIXED MODELS

- Our ways of evaluating models using an explanatory framework are limited when
 - we have no good way of estimating effective parameters
 - we have high degrees of collinearity
 - we have small sample sizes
 - the latter two can lead to unstable models
- Regularisation can improve the stability of a model
 - which comes at the cost of adding bias to the model

Learning goals and outline

Explanation versus prediction

- 1) Understanding how error can be decomposed into bias and variance
- 2) Understanding when penalised regression may be helpful
- 3) Understanding how models can be evaluated by out-of-sample testing

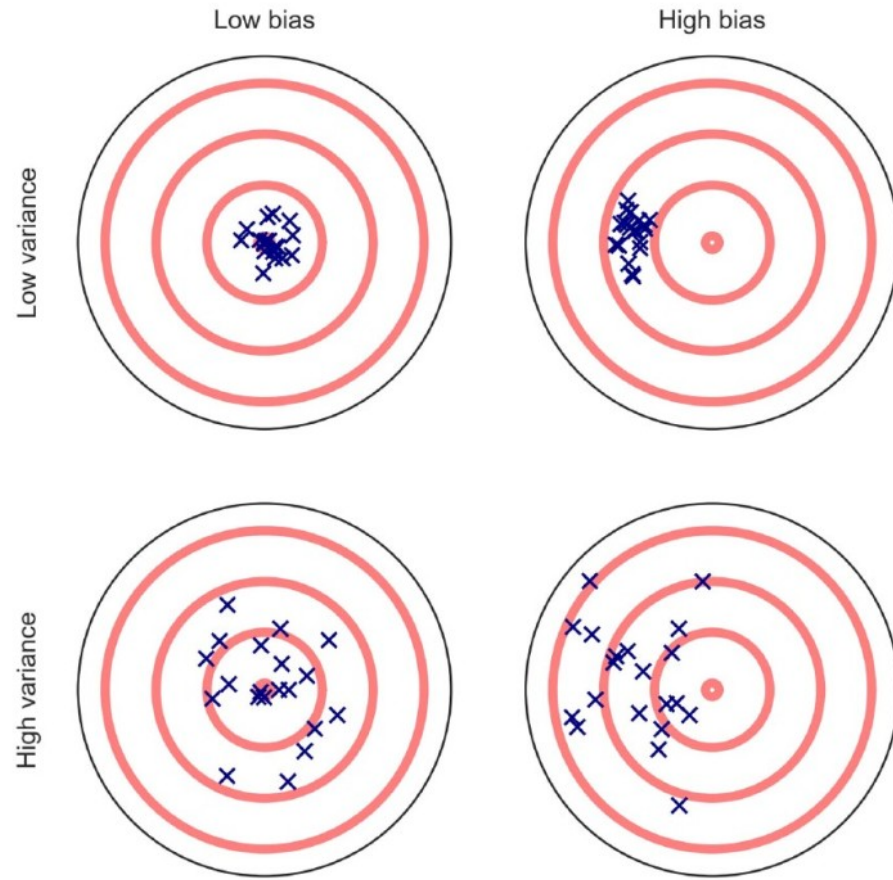


Fig. 2. An estimator's predictions can deviate from the desired outcome (or true scores) in two ways. First, the predictions may display a systematic tendency (or *bias*) to deviate from the central tendency of the true scores (compare right panels with left panels). Second, the predictions may show a high degree of *variance*, or imprecision (compare bottom panels with top panels).

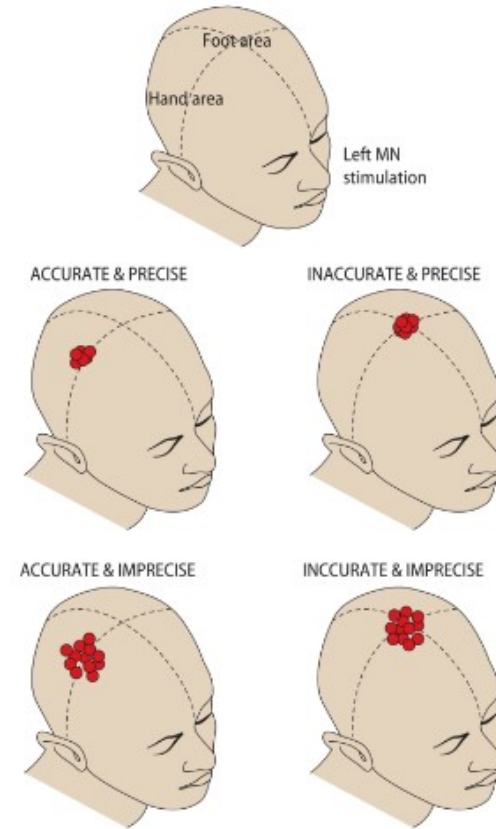


FIGURE 3.7. Accuracy versus precision. A schematic illustration of the differences between accuracy and precision of source localization. After left median-nerve stimulation, activations is expected in the right-hemisphere hand region of the primary somatosensory cortex. The foot area is shown at the top of the head. See text for further explanation.

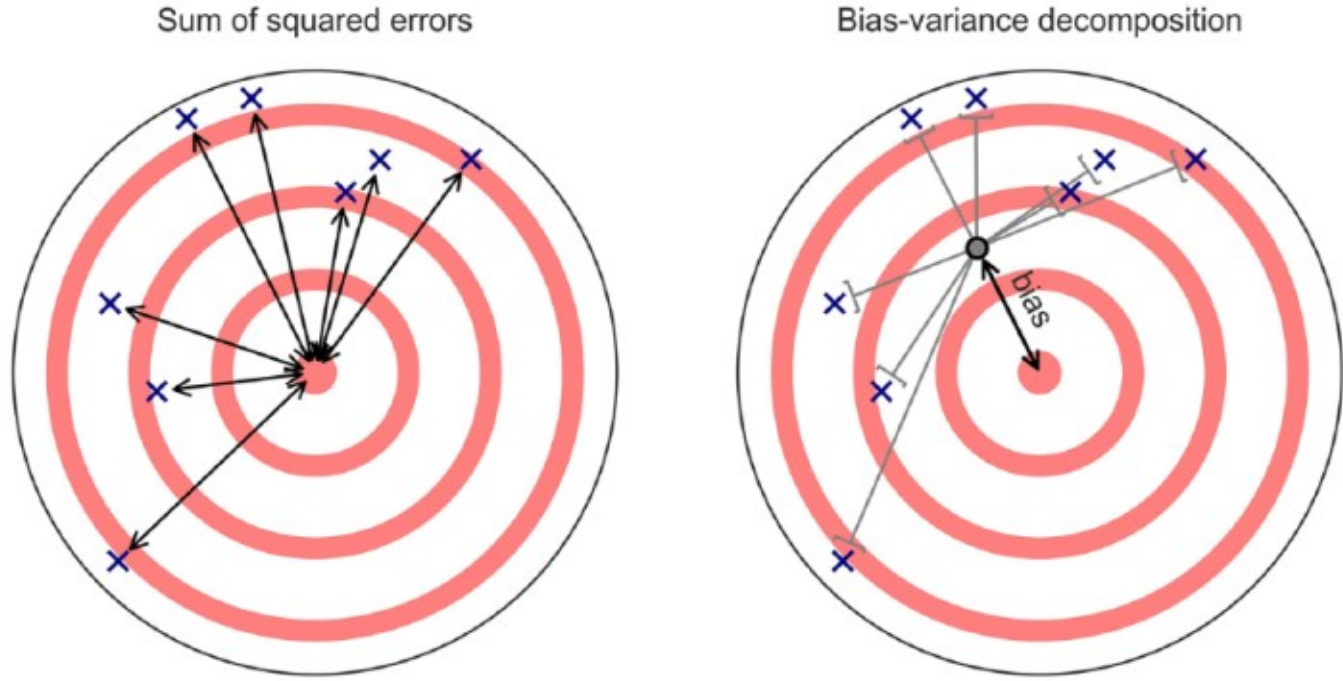


Fig. 3. Schematic illustration of the bias-variance decomposition. (Left) Under the classical error model, prediction error is defined as the sum of squared differences between true scores and observed scores (black lines). (Right) The bias-variance decomposition partitions the total sum of squared errors into two separate components: a bias term that captures a model's systematic tendency to deviate from the true scores in a predictable way (black line) and a variance term that represents the deviations of the individual observations from the model's expected prediction (gray lines).

Bias-variance decomposition

$$MSE = E[(f(x) - \hat{f}(x))^2] = bias(\hat{f}(x))^2 + var(\hat{f}(x)) + \sigma^2$$

$$bias(\hat{f}(x)) = E[\hat{f}(x)] - f(x)$$

MSE : mean squared error

$E[\]$: expected value (often the mean)

$f(x)$: true underlying function

$\hat{f}(x)$: estimated underlying function

$$MSE = E[(f(x) - \hat{f}(x))^2] = bias(\hat{f}(x))^2 + var(\hat{f}(x)) + \sigma^2$$

$$bias(\hat{f}(x)) = E[\hat{f}(x)] - f(x)$$

```
simulate.bias.and.var <- function(formula, fx, x0, x.range, n.sims, sigma)
{
  n.obs <- length(x.range)
  predictions <- matrix(data=NA, nrow=n.obs, ncol=n.sims)
  for(n.sim in 1:n.sims)
  {
    fx.noise <- fx(x.range) + rnorm(n=n.obs, sd=sigma)
    data <- data.frame(x=x.range, y=fx.noise)
    model <- lm(formula, data=data)
    fx.hat <- fitted(model)
    predictions[, n.sim] <- fx.hat
  }

  x.index <- which(x.range == x0)
  eps <- rnorm(n.sims, sd=sigma)
  MSE.estimated <- mean((predictions[x.index, ] - fx(x.range)[x.index] +
    eps)^2)
  bias <- mean(predictions[x.index, ]) - fx(x.range)[x.index]
  variance <- var(predictions[x.index, ])
  irreducible.error <- var(eps)
  MSE.theoretical <- bias^2 + variance + irreducible.error

  plot(y ~ x, data=data, main='Simulated data (Final draw)')
  lines(x.range, predict(model, newdata=data.frame(x=x.range)))
  print('Difference between estimated and theoretical MSE')
  print(abs(MSE.estimated - MSE.theoretical))
  return(c(bias^2, variance, irreducible.error))
}
```

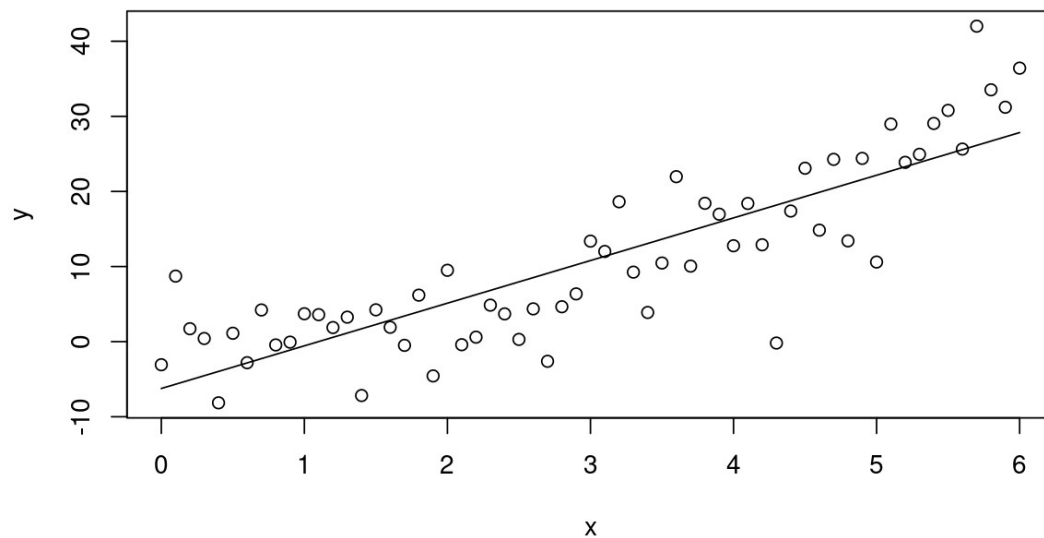
```
formulas <- c(
  y ~ x,
  y ~ I(x^2) + x,
  y ~ I(x^3) + I(x^2) + x,
  y ~ I(x^4) + I(x^3) + I(x^2) + x,
  y ~ I(x^5) + I(x^4) + I(x^3) + I(x^2) + x
)
```

```
for(formula in formulas)
{
  error <- simulate.bias.and.var(formula=formula, fx=function(x) x^2, x0=3,
                                x.range=seq(0, 6, 0.1),
                                n.sims=1000, sigma=5)

  errors <- rbind(errors, error)
}
```

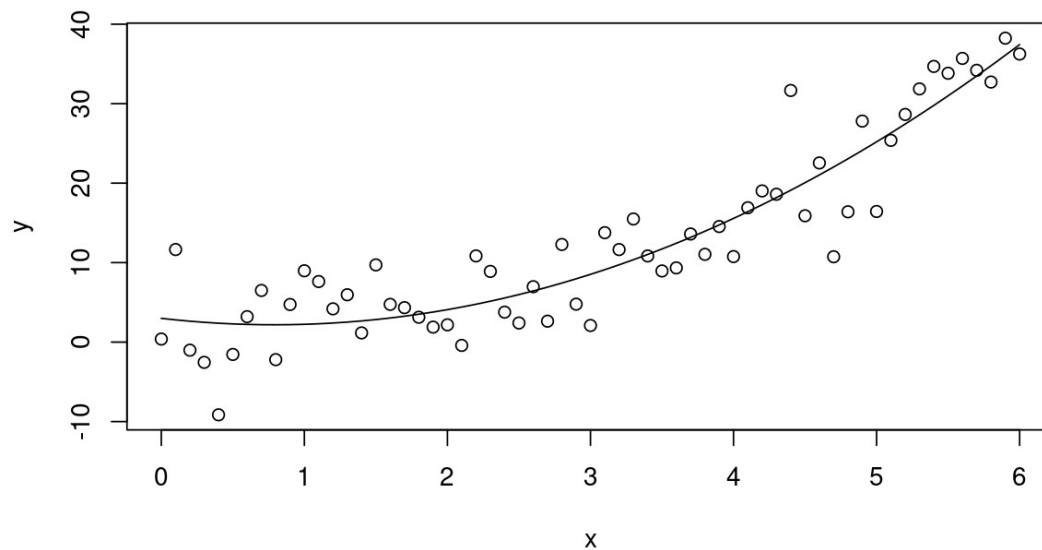
$$y \sim x$$

Simulated data (Final draw)



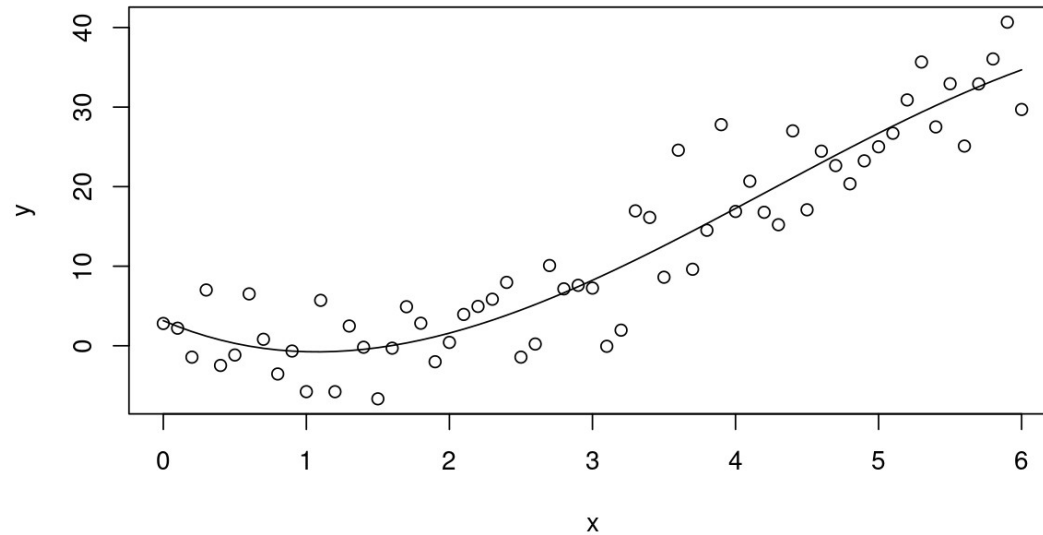
$$y \sim x^2 + x$$

Simulated data (Final draw)



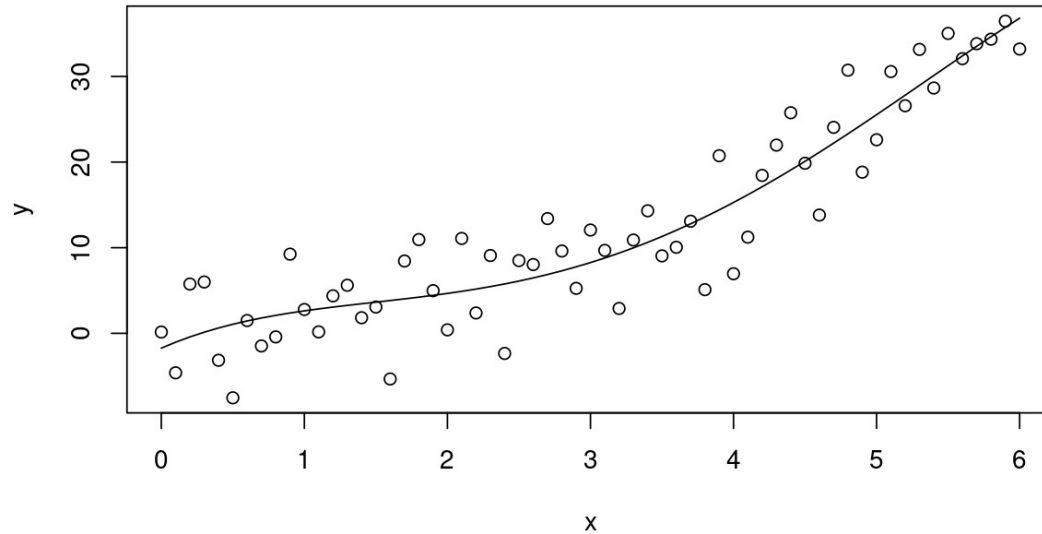
$$y \sim x^3 + x^2 + x$$

Simulated data (Final draw)



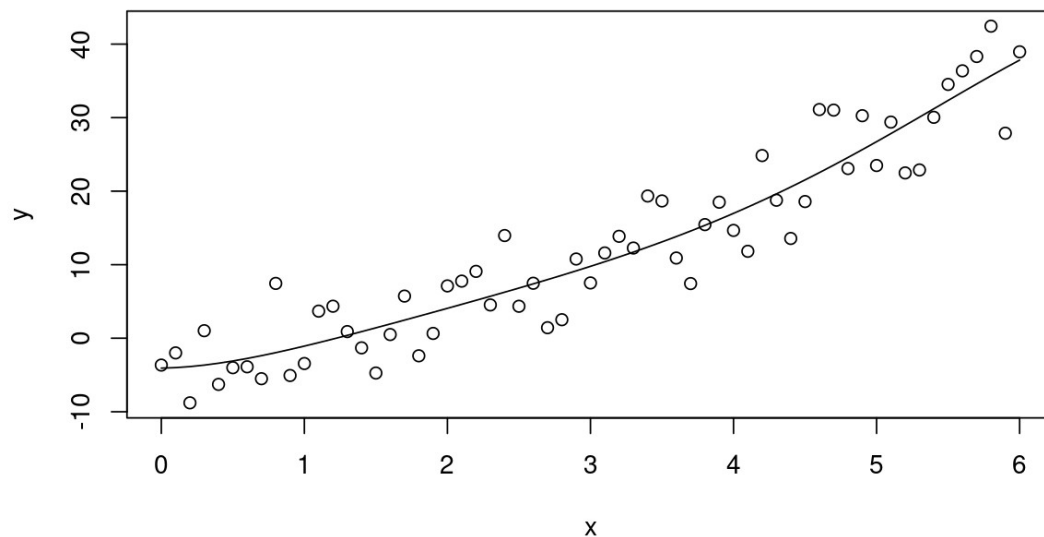
$$y \sim x^4 + x^3 + x^2 + x$$

Simulated data (Final draw)



$$y \sim x^5 + x^4 + x^3 + x^2 + x$$

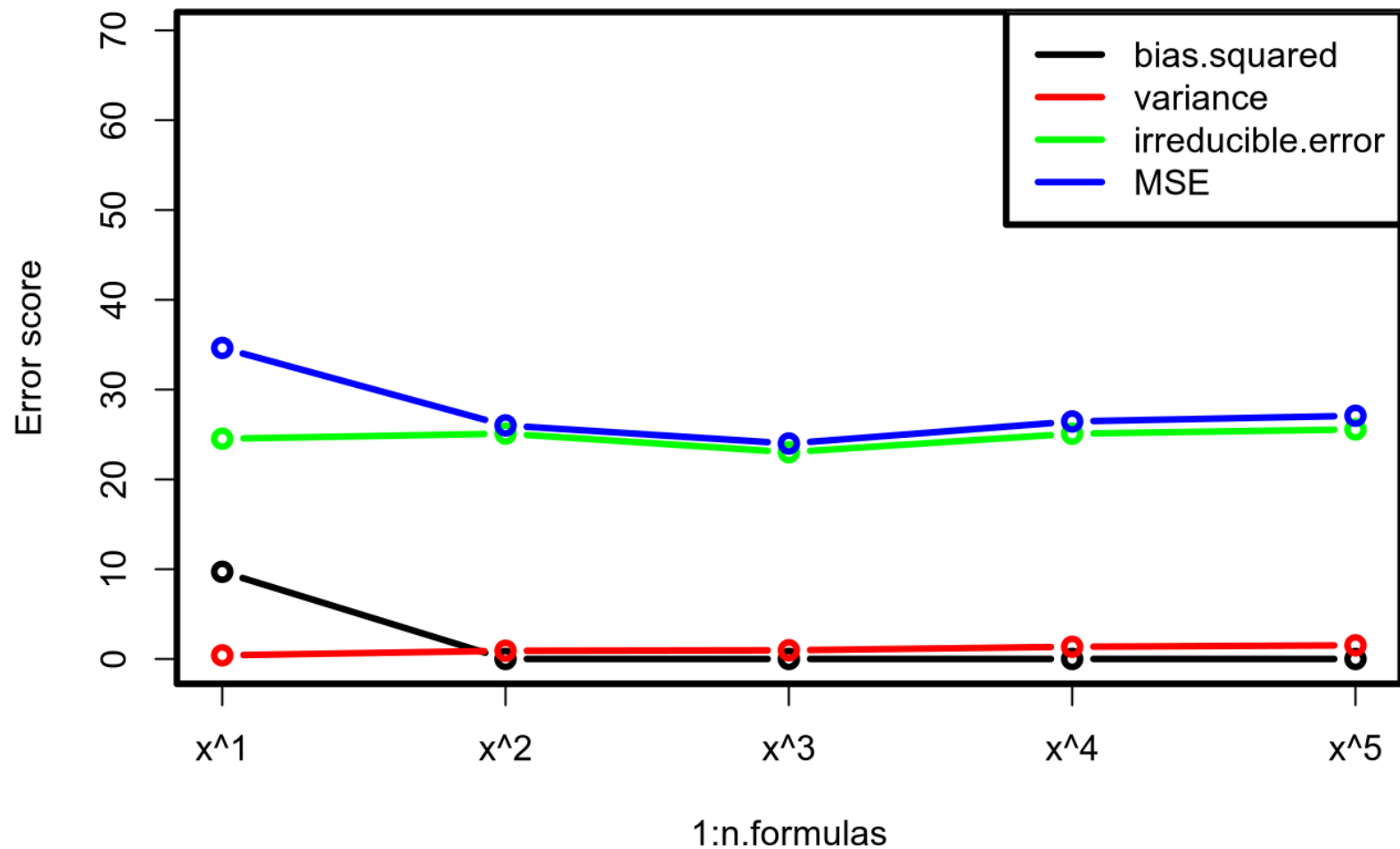
Simulated data (Final draw)



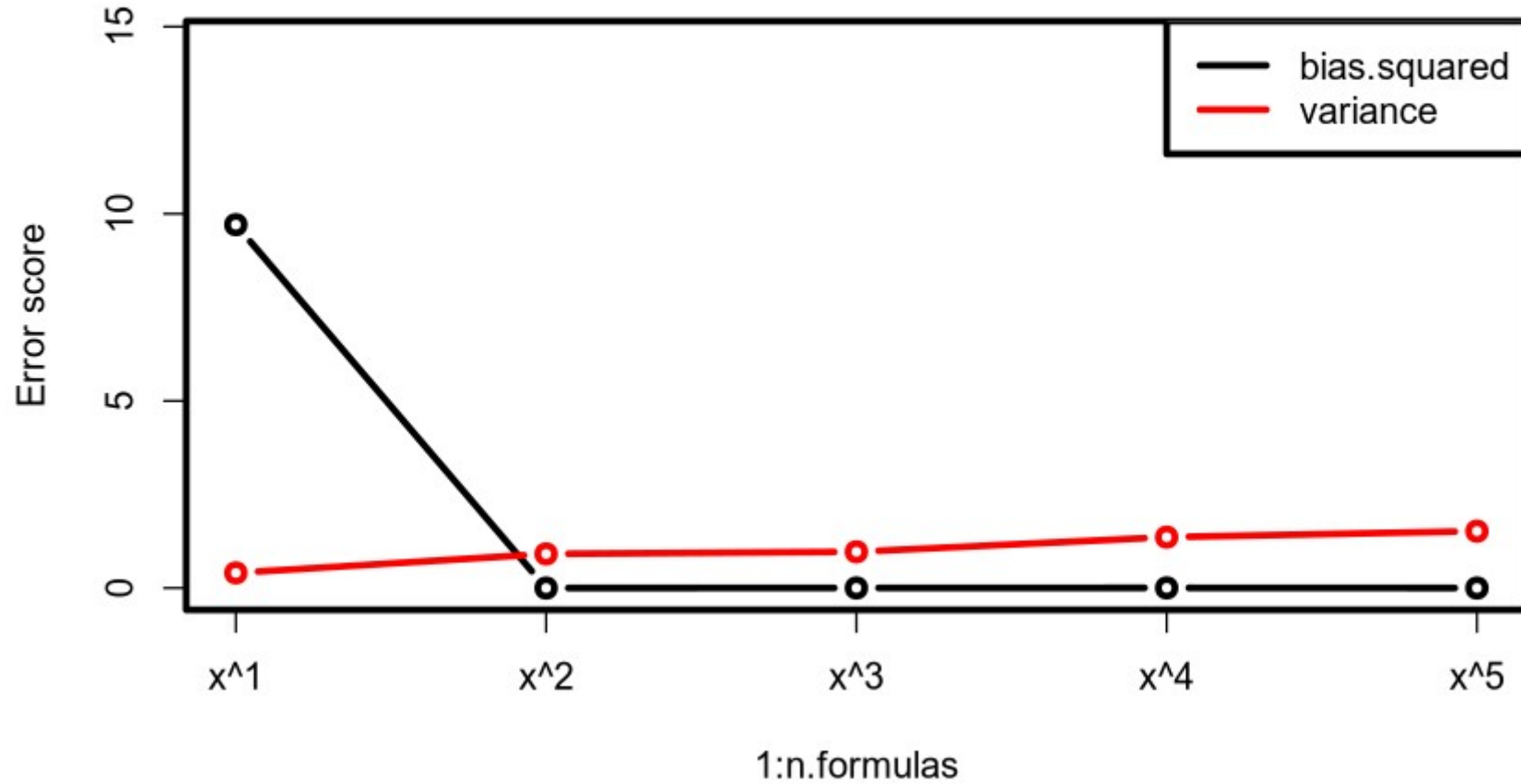
$$x_0 = 3$$

	##		bias.squared	variance	irreducible.error
$y \sim x$	##	1	9.708566e+00	0.4044735	24.52602
$y \sim x^2 \dots$	##	2	3.096235e-05	0.9119171	25.09829
$y \sim x^3 \dots$	##	3	1.852917e-03	0.9680085	23.02820
$y \sim x^4 \dots$	##	4	3.756766e-03	1.3596791	25.08635
$y \sim x^5 \dots$	##	5	1.799701e-04	1.5190037	25.56529

Models evaluated at $x_0 = 3$



Models evaluated at $x_0 = 3$

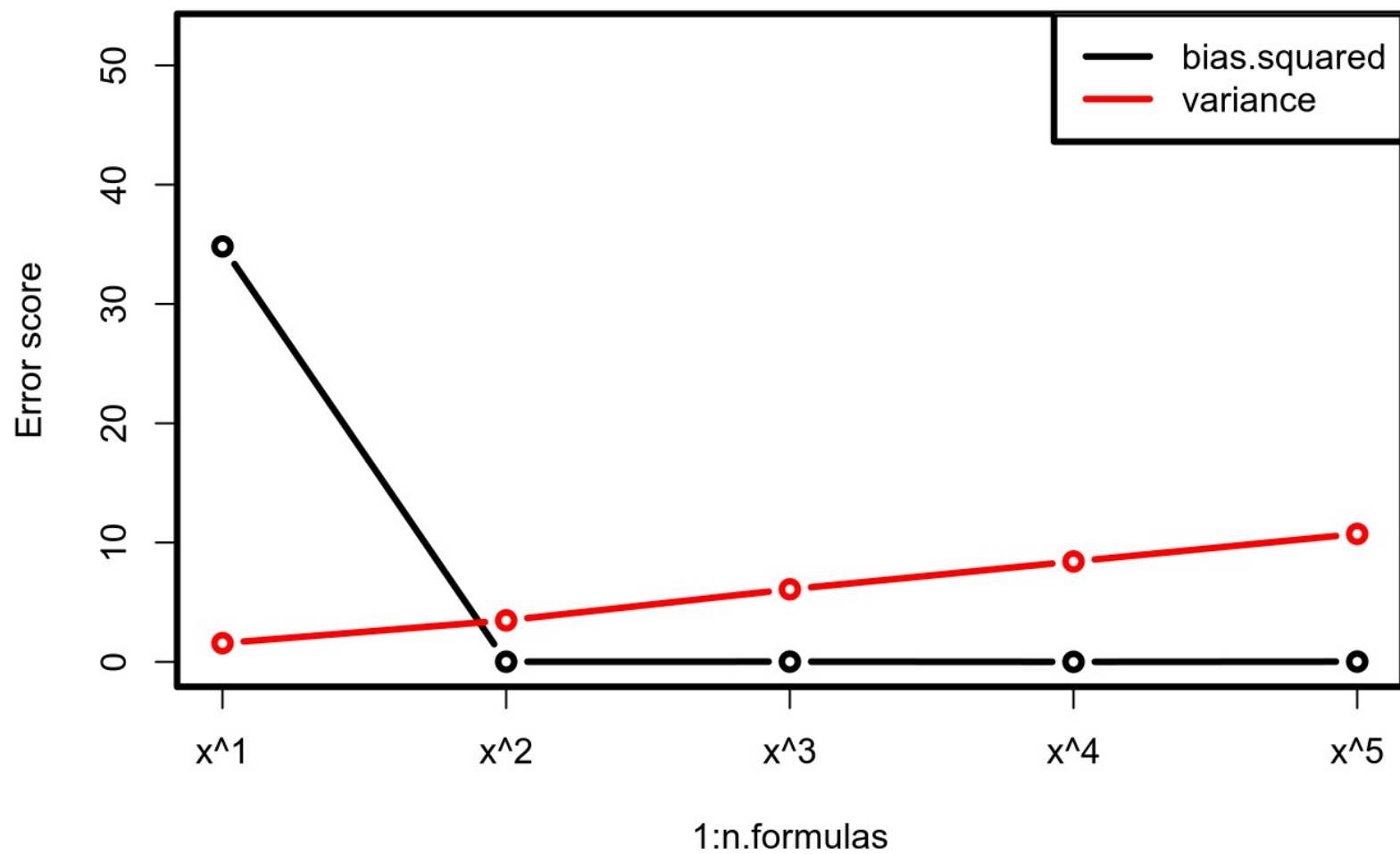


Discussion points:

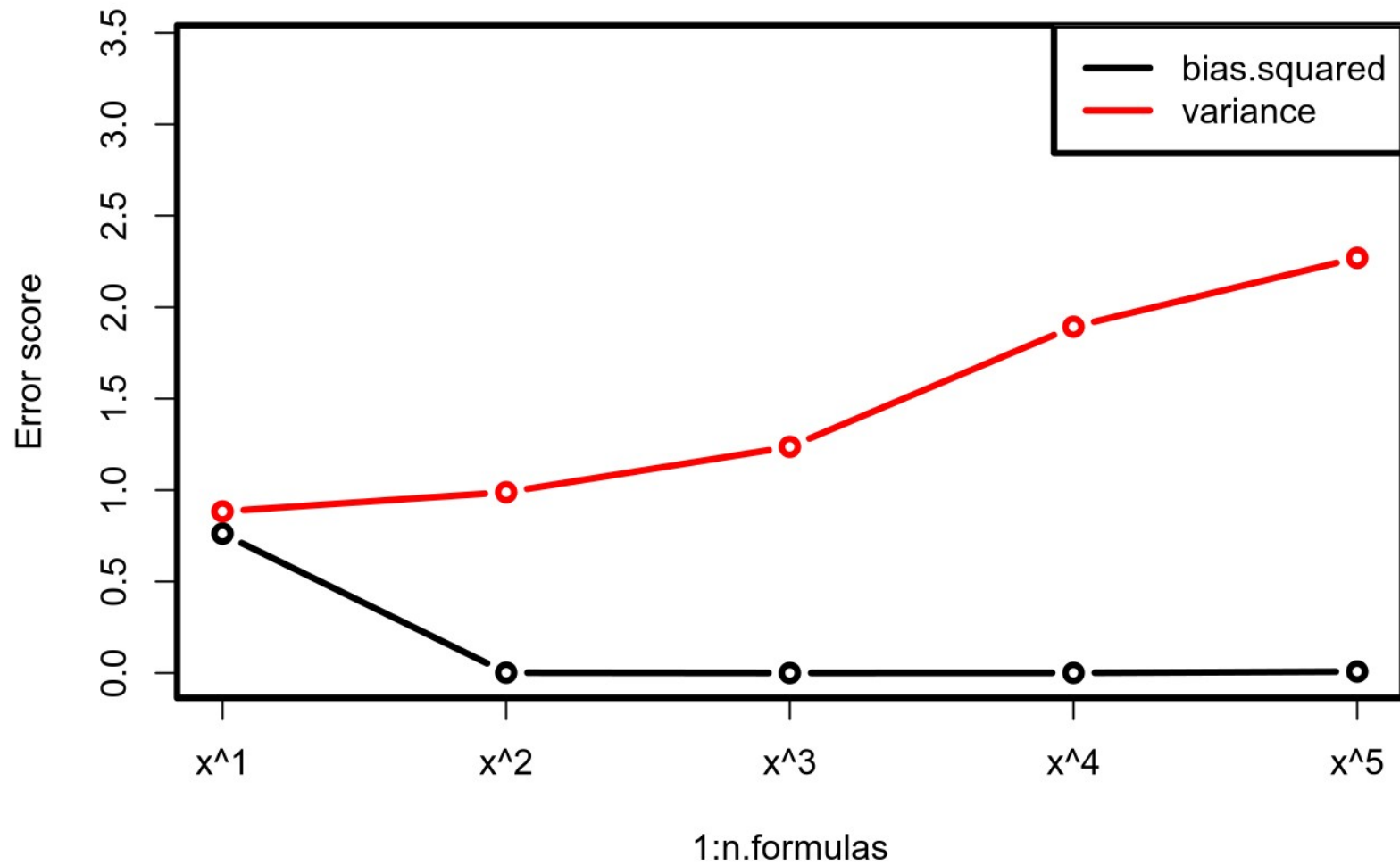
What happens to variance when x_0 is increased?
(say to 6)

What happens to variance when x_0 is decreased?
(say to 1)

Models evaluated at $x_0 = 6$



Models evaluated at $x_0 = 1$



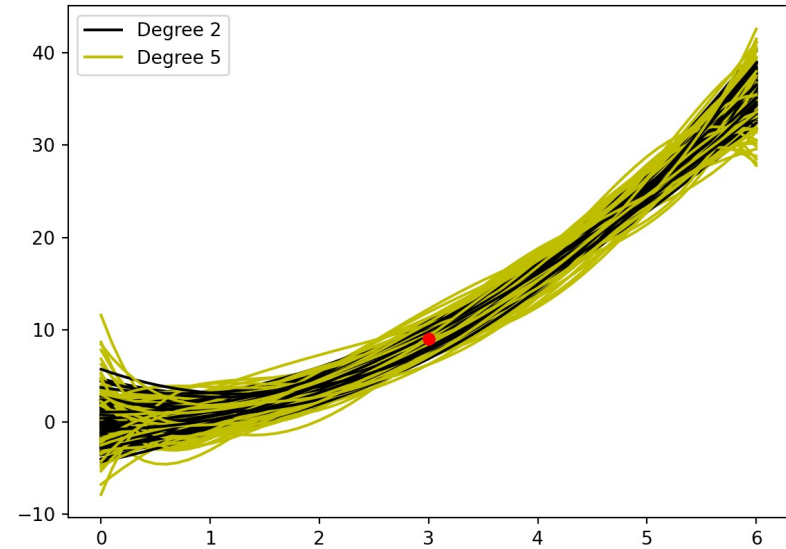
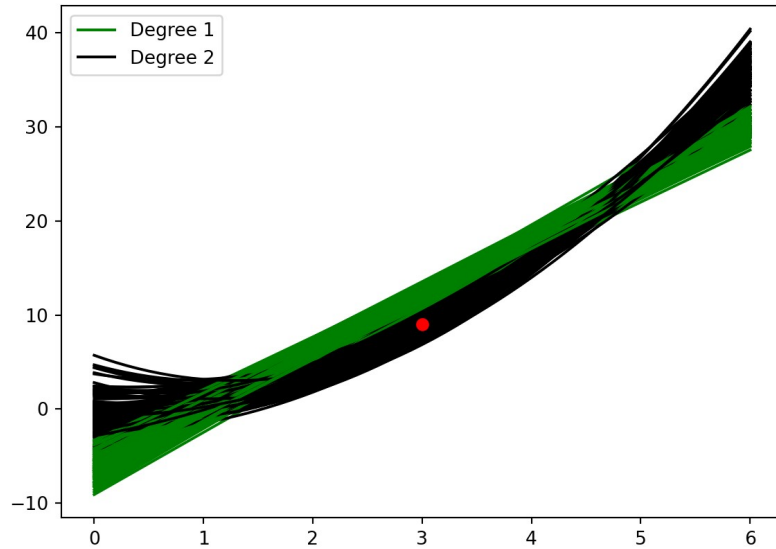
$x_0 = 1$

$x_0 = 3$

$x_0 = 6$

	bias.squared	variance	bias.squared	variance	bias.squared	variance
1	0.7620209937	0.8835371	9.708566e+00	0.4044735	34.821206760	1.565046
2	0.0017383448	0.9885182	3.096235e-05	0.9119171	0.012470904	3.474977
3	0.0001546774	1.2368800	1.852917e-03	0.9680085	0.018714117	6.082202
4	0.0007538773	1.8933452	3.756766e-03	1.3596791	0.002293859	8.414034
5	0.0078616986	2.2698603	1.799701e-04	1.5190037	0.019602168	10.724150

Simulations for polynomials of degree 1, 2 and 5 at $x_0 = 3$



OUT OF SAMPLE TESTING as a way of evaluating models

Using mean squared error as a
measure of predictive power

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$


```

predict.out.of.sample <- function(x, fx, formula, sigma, n.sims)
{
  n.obs <- length(x)
  MSE.trains <- numeric(n.sims)
  MSE.tests <- numeric(n.sims)
  for(n.sim in 1:n.sims)
  {
    fx.train <- fx(x) + rnorm(n.obs, sd=sigma)
    fx.test <- fx(x) + rnorm(n.obs, sd=sigma)

    data.train <- data.frame(x=x, y=fx.train)
    data.test <- data.frame(x=x, y=fx.test)

    model <- lm(formula, data=data.train)
    fx.hat.train <- fitted(model)
    fx.hat.test <- predict(model, data.test)
    MSE.trains[n.sim] <- 1/n.obs * sum((fx.train - fx.hat.train)^2)
    MSE.tests[n.sim] <- 1/n.obs * sum((fx.test - fx.hat.test)^2)
  }
  print(formula)
  print(paste('MSE train:', round(mean(MSE.trains), 2)))
  print(paste('MSE test:', round(mean(MSE.tests), 2)))
  return(c(mean(MSE.trains), mean(MSE.tests)))
}

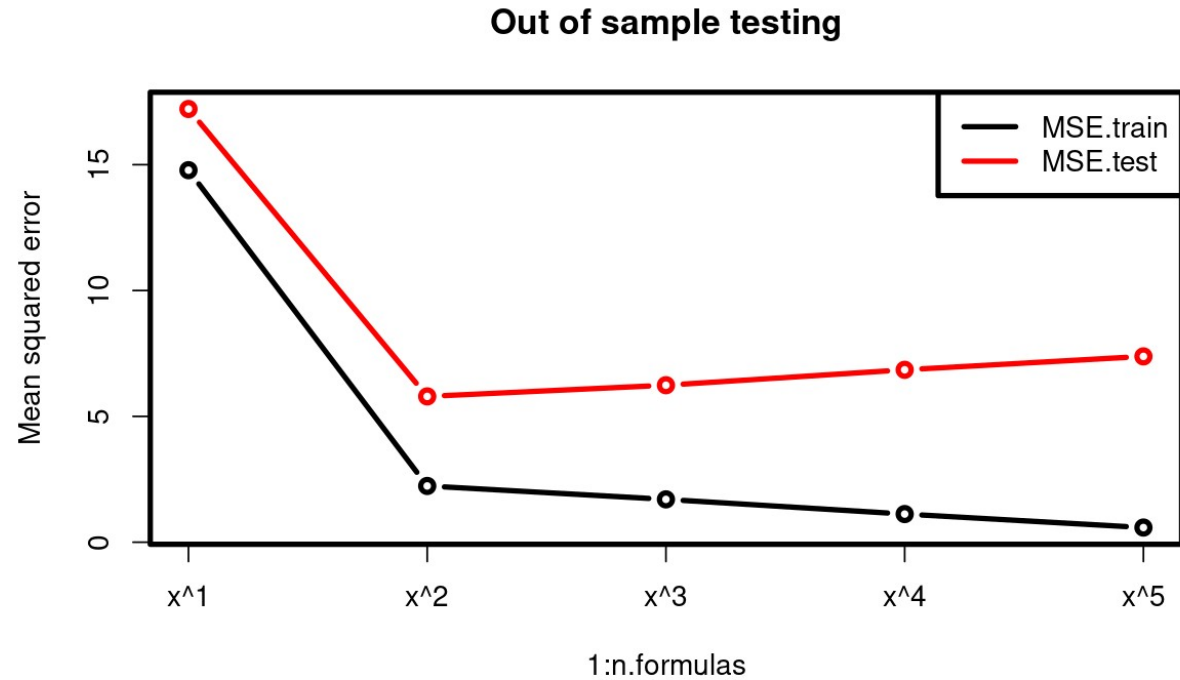
```

```

formulas <- c(
  y ~ x,
  y ~ I(x^2) + x,
  y ~ I(x^3) + I(x^2) + x,
  y ~ I(x^4) + I(x^3) + I(x^2) + x,
  y ~ I(x^5) + I(x^4) + I(x^3) + I(x^2) + x
)
n.formulas <- length(formulas)
MSEs <- matrix(data=NA, nrow=2, ncol=n.formulas)
for(formula.index in 1:n.formulas)
{
  formula <- formulas[[formula.index]]
  MSEs[, formula.index] <- predict.out.of.sample(x=seq(0, 6, 1),
                                                    fx=function(x) x^2,
                                                    formula=formula, sigma=2, n.sims=100)
}

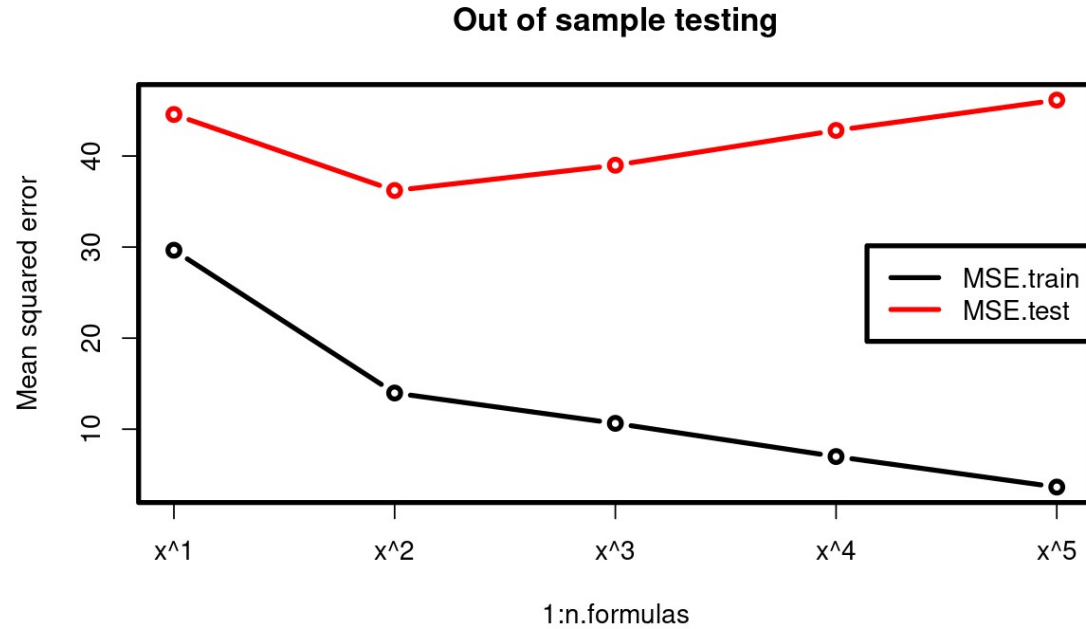
```

$\sigma=2$
 $n . obs=7$



$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

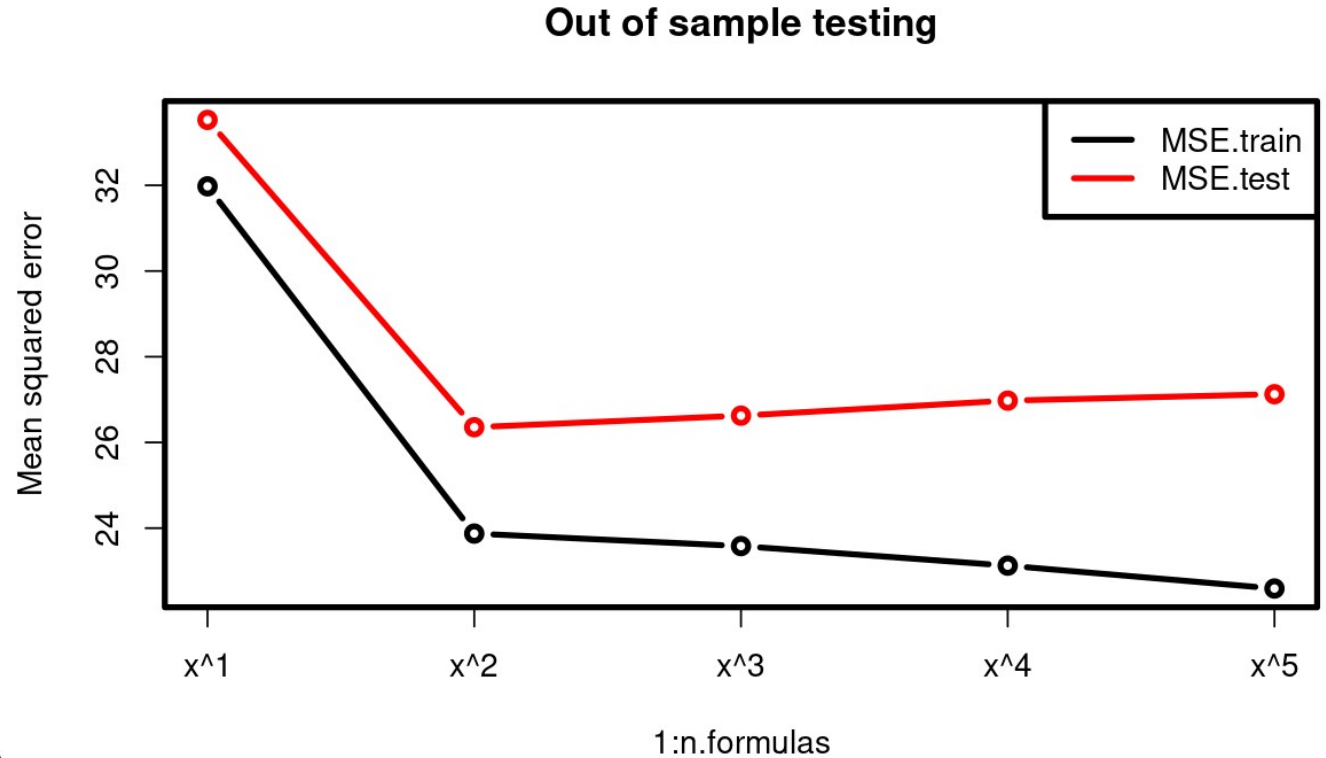
$\sigma=5$
 $n . obs=7$



$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

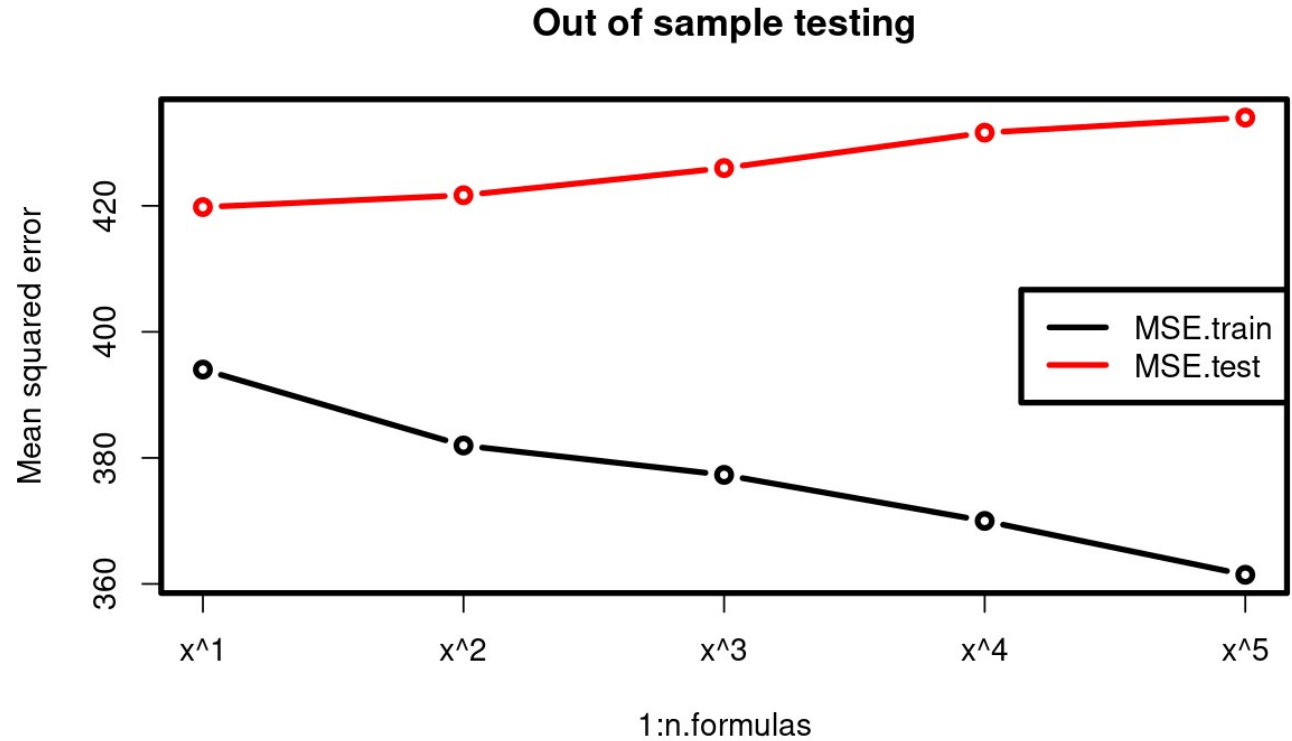
$\sigma=5$
 $n . obs=61$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

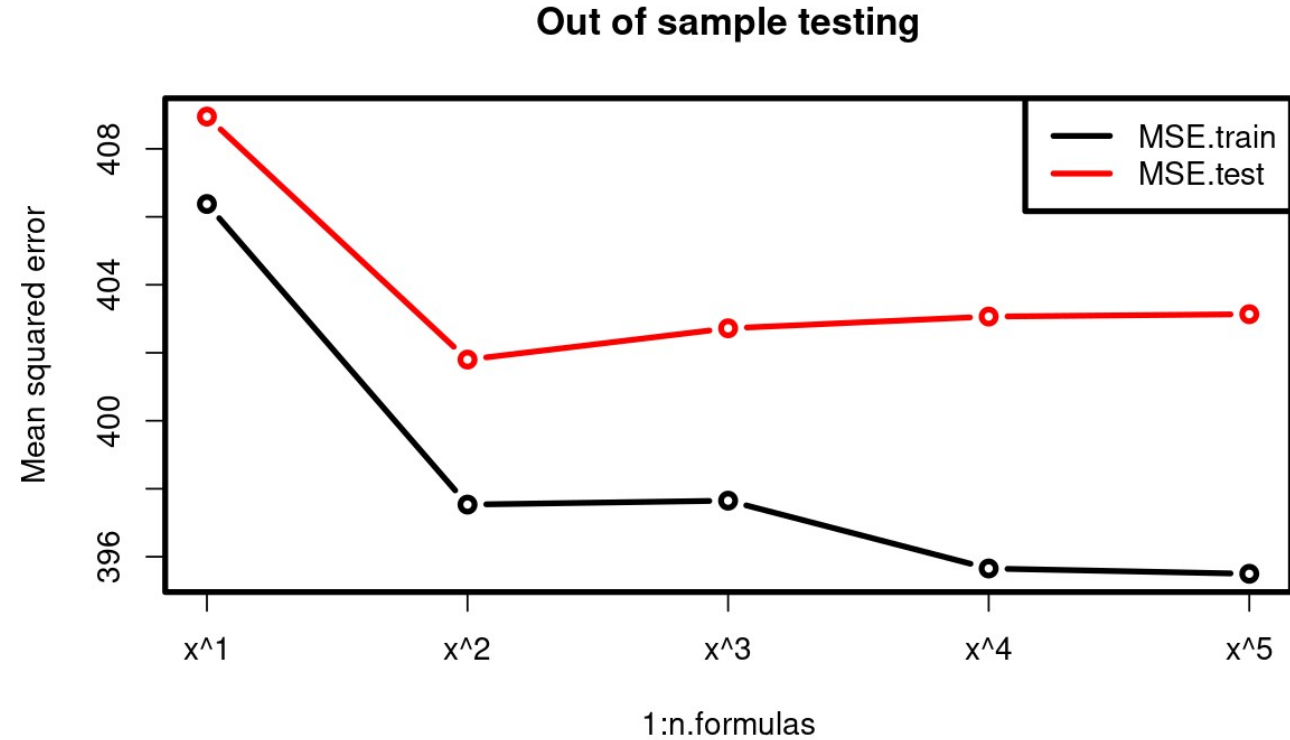


$\sigma=20$
 $n . obs=61$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



$\sigma=20$
 $n . obs=601$



$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Interim summary

So if greater variance is detrimental to prediction, we may be able improve prediction by introducing bias (and thereby reducing variance)

Penalised regression

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \text{ (minimise to obtain least squares solution)}$$

$$\text{lasso regression: } \text{RSS} + \lambda \sum_{j=1}^p |\beta_j| \text{ (minimise this sum)}$$

$$\text{ridge regression: } \text{RSS} + \lambda \sum_{j=1}^p (\beta_j^2) \text{ (minimise this sum)}$$

n : number of observations

p : number of predictor variables

λ : a constant

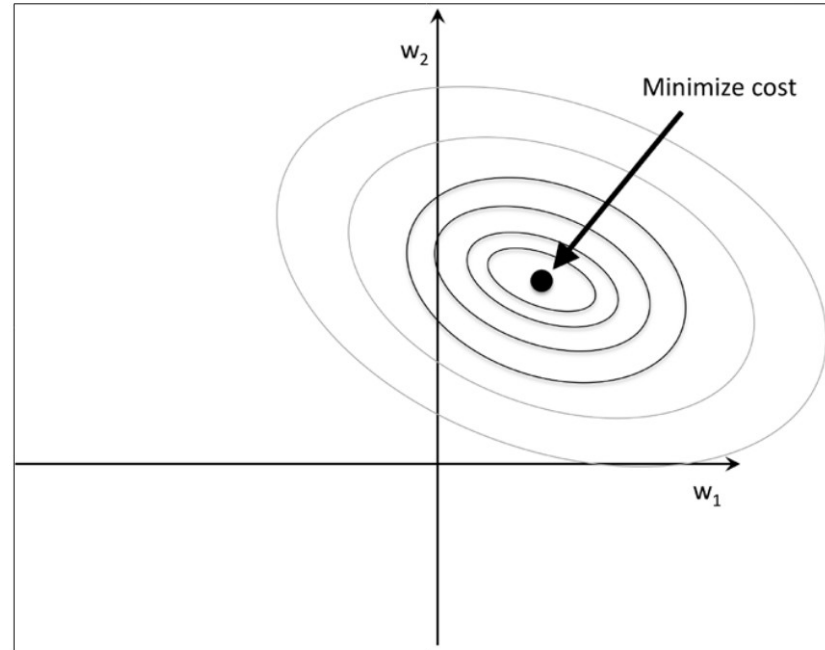
Group discussion

In each case: what happens when?

1. λ increases?
2. λ decreases?
3. λ is 0?
4. λ goes towards infinity?

Least squares

$$J(w) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



p. 113; Raschka S (2015) Python Machine Learning. Packt Publishing Ltd

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L2 regularization, ridge

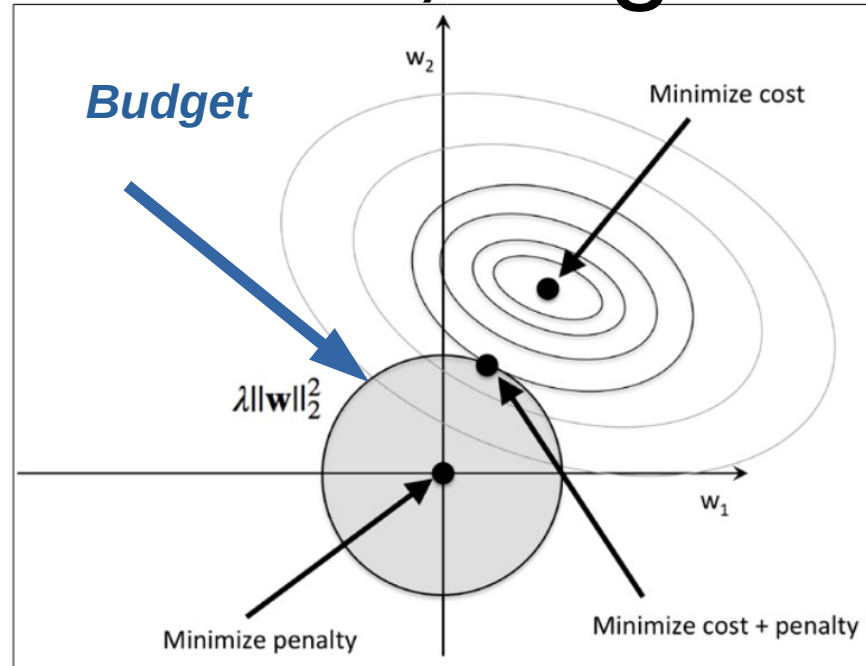
Why is the *budget* round?

Compare with a circle centred at (0,0)

$$x^2 + y^2 = r^2$$

$$w_1^2 + w_2^2 = r^2$$

$$\|w\|_2 = \sqrt{(w_1^2 + w_2^2)}$$



(p. 114: Raschka, 2015)

$$J(w)_{Ridge} = \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2 + \lambda \|w\|_2^2$$

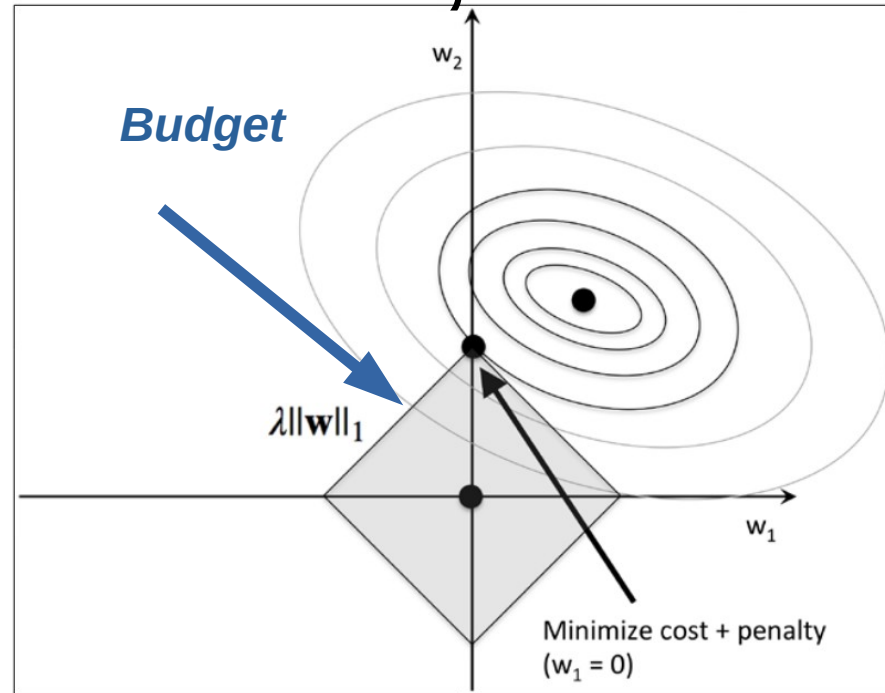
L1 regularization, lasso

Why is the *budget* square?

$$\|w\|_1 = |w_1| + |w_2|$$

if $w_1 = \max(w_1)$ then $w_2 = 0$

if $w_2 = \max(w_2)$ then $w_1 = 0$



(p. 115: Raschka, 2015)

$$J(w)_{LASSO} = \sum_{i=1}^n \left(y^{(i)} - \hat{y}^{(i)} \right)^2 + \lambda \|w\|_1$$

Out-of-sample as validity check



```
mtcars.1 <- mtcars[1:10, ]
```



Out-of-sample as validity check

Call:

```
lm(formula = hp ~ mpg + wt + drat + qsec, data = mtcars.1)
```

Coefficients:

(Intercept)	mpg	wt	drat	qsec
414.541	-13.638	12.753	11.263	-5.042

Call:

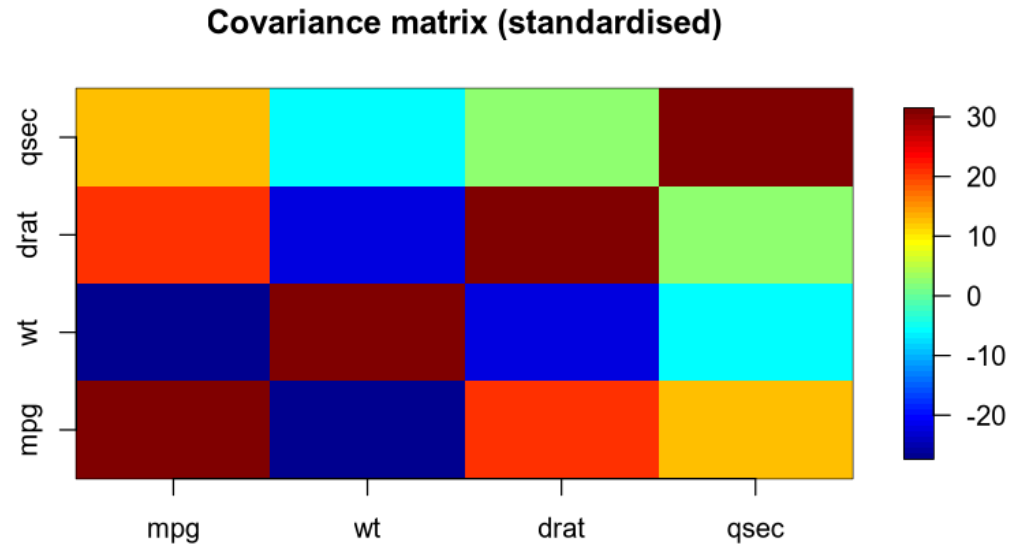
```
lm(formula = hp ~ mpg + wt + drat + qsec, data = mtcars)
```

Coefficients:

(Intercept)	mpg	wt	drat	qsec
473.779	-2.877	26.037	4.819	-20.751

Collinearity

MTCARS



Suddenly, someone shows up with



Let's check
our model

```
mtcars.2 <- mtcars[11:32, ]
```



```

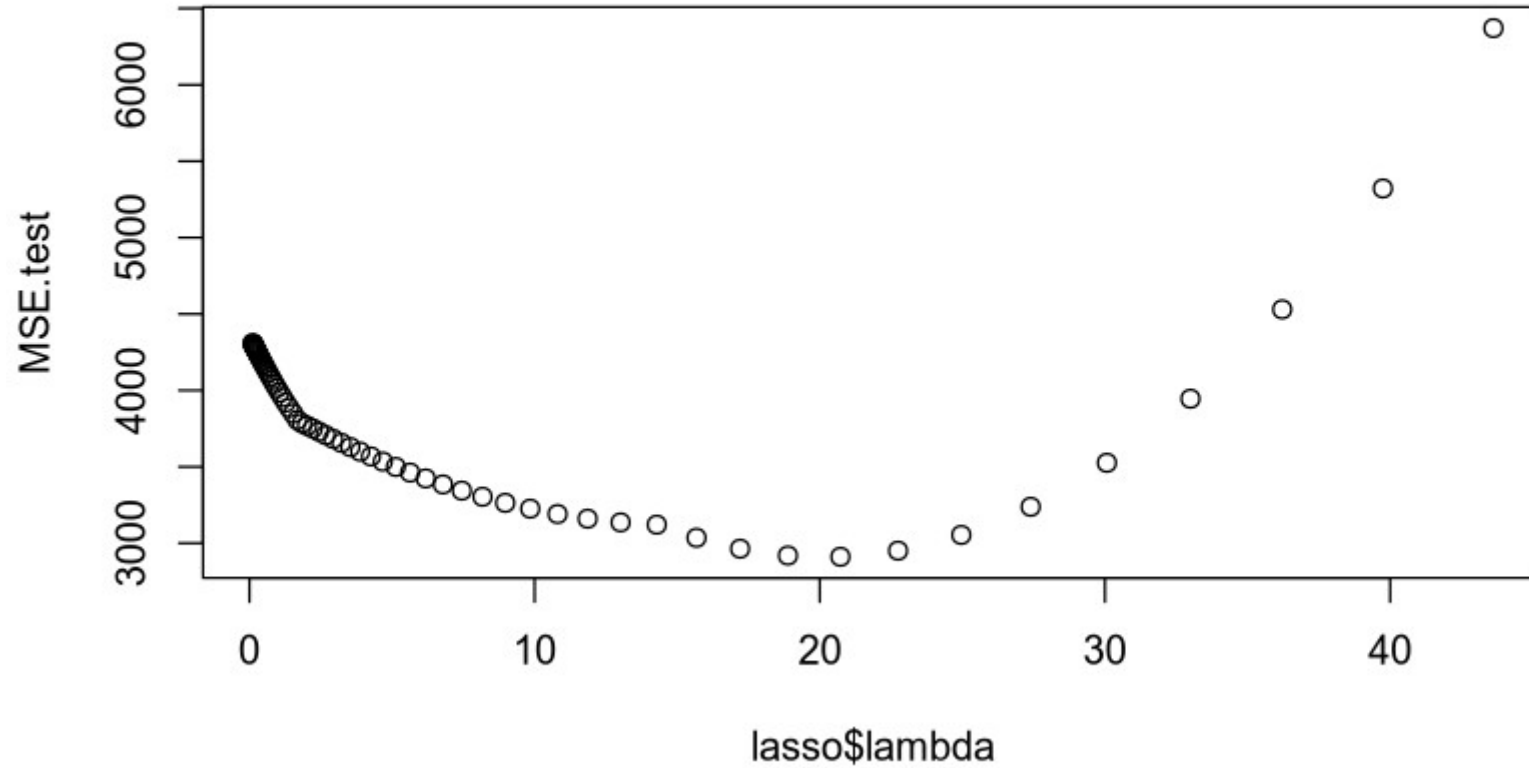
rm(list =ls())
library(glmnet)
y.train <- as.matrix(mtcars[1:10, 4]) ## hp
x.train <- as.matrix(mtcars[1:10, c(1, 6, 5, 7)]) # mpg, wt, drat, qsec
y.test <- as.matrix(mtcars[11:32, 4]) ## hp
x.test <- as.matrix(mtcars[11:32, c(1, 6, 5, 7)])# mpg, wt, drat, qsec

lasso <- glmnet(x=x.train, y=y.train, alpha=1) ## get RSS and penalty in
n.models <- lasso$dim[2]
MSE.train <- numeric(n.models)
penalty.train <- numeric(n.models)
MSE.test <- numeric(n.models)
for(model.index in 1:n.models)
{
  betas <- c(lasso$a0[model.index], lasso$beta[, model.index])
  X.train <- cbind(1, x.train)
  X.test <- cbind(1, x.test)
  y.hat.train <- X.train %*% betas
  y.hat.test <- X.test %*% betas
  MSE.train[model.index] <- 1 / length(y.train) * sum((y.train - y.hat.train)^2)
  penalty.train[model.index] <- sum(abs(lasso$beta[, model.index]))
  MSE.test[model.index] <- 1/length(y.test) * sum((y.test - y.hat.test)^2)
}

plot(lasso$lambda, MSE.test)

```

Adding bias helps prediction



lambda <dbl>	MSE.train <dbl>	penalty.train <dbl>	MSE.test <dbl>
43.6220702	2382.3600	0.000000	6371.040
39.7468057	2059.2836	1.405142	5321.815
36.2158090	1791.0598	2.685456	4530.285
32.9984963	1568.3758	3.852030	3945.631
30.0670009	1383.4995	4.914968	3526.290
27.3959314	1230.0121	5.883478	3238.326
24.9621524	1102.5840	6.765949	3054.088
22.7445835	996.7911	7.570023	2951.093
20.7240174	908.9599	8.302665	2911.110
18.8829528	836.0409	8.970222	2919.395

```
glmnet(x = x.train, y = y.train, alpha = 1)
```

s8	mpg	wt	drat	qsec
291.925294	-8.302665	0.000000	0.000000	0.000000

Call:

```
lm(formula = hp ~ mpg + wt + drat + qsec + 1, data = mtcars[1:10, ])
```

Coefficients:

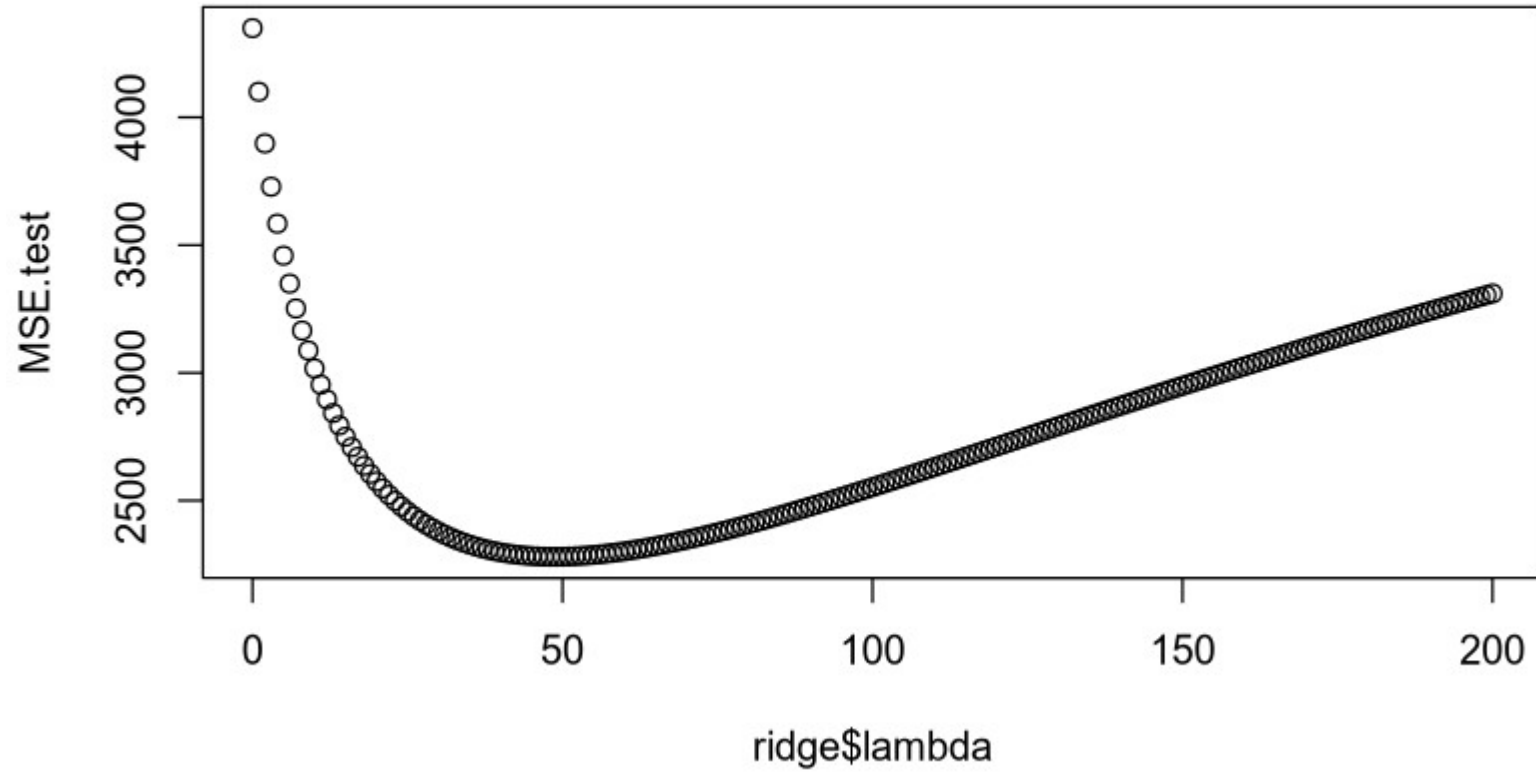
(Intercept)	mpg	wt	drat	qsec
414.541	-13.638	12.753	11.263	-5.042

```
## ridge
rm(list =ls())
library(glmnet)
y.train <- as.matrix(mtcars[1:10, 4]) ## hp
x.train <- as.matrix(mtcars[1:10, c(1, 6, 5, 7)]) # mpg, wt, drat, qsec
y.test <- as.matrix(mtcars[11:32, 4]) ## hp
x.test <- as.matrix(mtcars[11:32, c(1, 6, 5, 7)])# mpg, wt, drat, qsec

ridge <- glmnet(x=x.train, y=y.train, alpha=0, lambda=0:200)
n.models <- ridge$dim[2]
MSE.train <- numeric(n.models)
penalty.train <- numeric(n.models)
MSE.test <- numeric(n.models)
for(model.index in 1:n.models)
{
  betas <- c(ridge$a0[model.index], ridge$beta[, model.index])
  X.train <- cbind(1, x.train)
  X.test <- cbind(1, x.test)
  y.hat.train <- X.train %*% betas
  y.hat.test <- X.test %*% betas
  MSE.train[model.index] <- 1 / length(y.train) * sum((y.train - y.hat.train)^2)
  penalty.train[model.index] <- sum(ridge$beta[, model.index]^2)
  MSE.test[model.index] <- 1/length(y.test) * sum((y.test - y.hat.test)^2)
}

plot(ridge$lambda, MSE.test, main='Adding bias helps prediction')
```

Adding bias helps prediction



lambda <dbl>	MSE.train <dbl>	penalty.train <dbl>	MSE.test <dbl>
50	711.7847	409.7758	2280.286
49	705.3788	413.2714	2279.970
48	698.9470	416.7924	2280.049
47	692.4899	420.3382	2280.541
46	686.0082	423.9078	2281.469
45	679.5026	427.5003	2282.853
44	672.9740	431.1146	2284.718
43	666.4231	434.7496	2287.087
42	659.8510	438.4039	2289.987
41	653.2585	442.0760	2293.446

```
glmnet(x = x.train, y = y.train, alpha = 0, lambda = 0:200)
      s151      mpg      wt      drat      qsec
312.148253 -5.788459 17.325500 -7.093469 -5.410624
```

Call:

```
lm(formula = hp ~ mpg + wt + drat + qsec + 1, data =
mtcars[1:10, ])
```

Coefficients:

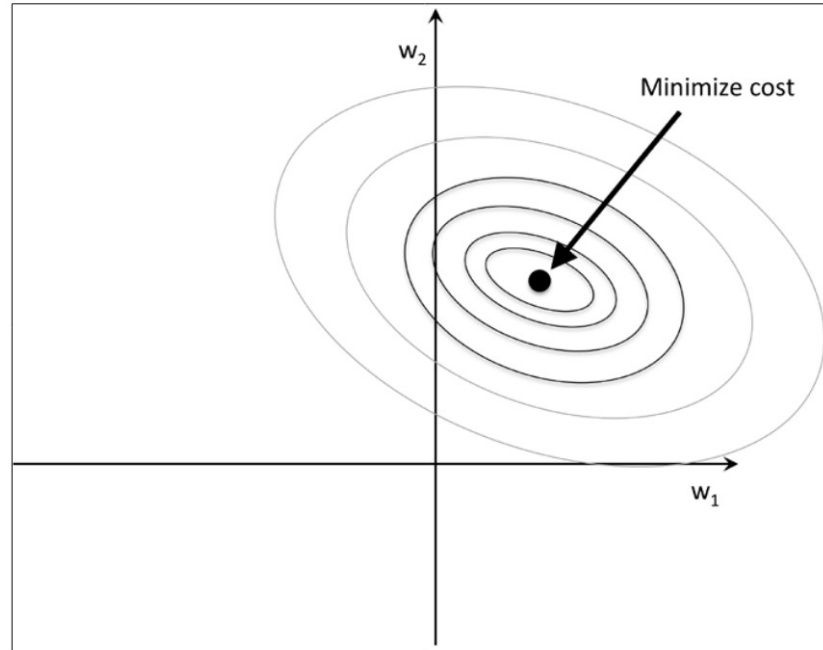
```
(Intercept)      mpg      wt      drat      qsec
  414.541    -13.638    12.753    11.263    -5.042
```


Adding bias to a model can increase stability of the model and can in turn increase prediction capability

Remember in linear regression MSE is a metric for prediction error

Least squares

$$J(w) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



p. 113; Raschka S (2015) Python Machine Learning. Packt Publishing Ltd

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L2 regularization, ridge

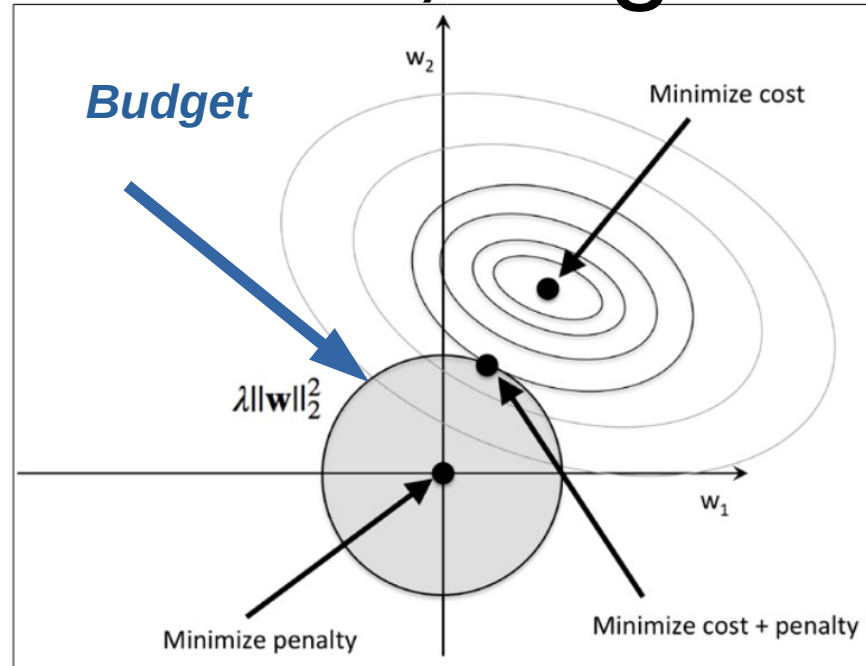
Why is the *budget* round?

Compare with a circle centred at (0,0)

$$x^2 + y^2 = r^2$$

$$w_1^2 + w_2^2 = r^2$$

$$\|w\|_2 = \sqrt{(w_1^2 + w_2^2)}$$



(p. 114: Raschka, 2015)

$$J(w)_{Ridge} = \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2 + \lambda \|w\|_2^2$$

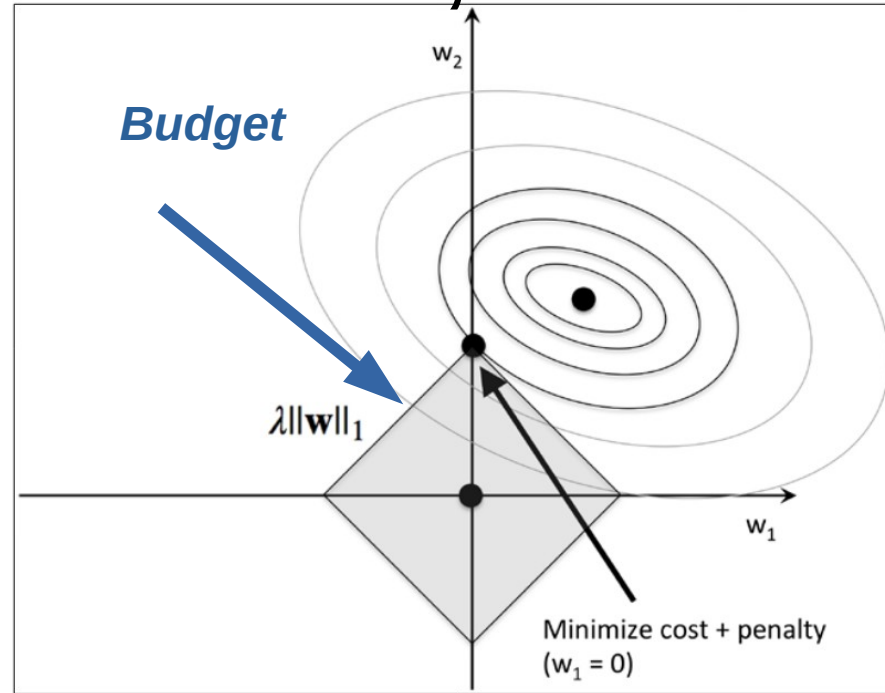
L1 regularization, lasso

Why is the *budget* square?

$$\|w\|_1 = |w_1| + |w_2|$$

if $w_1 = \max(w_1)$ then $w_2 = 0$

if $w_2 = \max(w_2)$ then $w_1 = 0$



(p. 115: Raschka, 2015)

$$J(w)_{LASSO} = \sum_{i=1}^n \left(y^{(i)} - \hat{y}^{(i)} \right)^2 + \lambda \|w\|_1$$

We can combine L1 and L2

JUST FOR COMPLETION

$$J(w)_{ElasticNet} = \sum_{i=1}^n \left(y^{(i)} - \hat{y}^{(i)} \right)^2 + \lambda_1 \sum_{j=1}^m w_j^2 + \lambda_2 \sum_{j=1}^m |w_j|$$

Summary

- Bias can be added in ways that improve prediction
 - it does this by removing collinearity
 - thereby making the model stable
 - and more generalisable
- This is one backbone of machine learning

Learning goals and outline

Explanation versus prediction

- 1) Understanding how error can be decomposed into bias and variance
- 2) Understanding when penalised regression may be helpful
- 3) Understanding how models can be evaluated by out-of-sample testing

The course plan

Week 37:

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model (Lau)

Week 38:

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects? (Lydia)

Week 39:

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous? (Lau)

Week 40:

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review (Lydia)

Week 41:

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models (Lydia)

Week 42:

no teaching (go fetch potatoes instead)

Week 43:

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running (Lau)

Week 44:

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses (Lau)

Week 45:

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review (Lydia: To be moved)

Week 46:

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data? (Lau)

Week 47:

Lesson 9: Neural networks

Class 9: Code review (Lau)

Week 48:

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource) (Lydia)

Week 49:

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning (Lydia)

Week 50:

Student presentations:

Class 12: Work on portfolios: Ask anything! (Lydia & Lau)

Next time

- Introduction to classification
 - The Perceptron
 - ADALine
- Linear regression in machine learning
 - Looking at big(ger) scale data

Reading questions

- Chapter 1
 - What are the differences between supervised, unsupervised and reinforcement learning?
 - What is the difference between classification and regression?
 - What is dimensionality reduction?
 - Is it similar to regularisation?
- Chapter 2
 - How does the w vector and x matrix relate to what we know as X and β ?
 - What is the difference between a training data set and a test data set?
 - What is gradient descent?
 - What is a quantizer?